⭐⭐⭐⭐⭐ 1. Read, decode and make sense of statements, questions, tasks, objects or images (Item 4)

Most central

- Students must correctly understand that:
  - "One full rotation in 40 minutes" implies **constant speed**
  - "Half an hour" equals **30 minutes**
  - The starting point is **P**
  - The answer options refer to **specific positions on the diagram**
- Without accurate interpretation of the text and diagram, the position cannot be determined

👉 This reflects the ability to **construct a situation model from verbal and visual information**

⭐⭐⭐⭐⭐ 2. Recognise mathematical structure (including regularities, relationships, and patterns) (Item 5)

Very important

- Students need to recognise the underlying structure:
  - Time is **directly proportional** to the angle rotated
  - 40 minutes → 360°
  - 30 minutes → $\frac{30}{40} = \frac{3}{4}$ of a full rotation
- This is a **cyclical and proportional structure**

👉 This demonstrates recognition of **periodic motion and proportional relationships**

⭐⭐⭐⭐ 3. Recognise aspects of a problem that correspond with known mathematical concepts, facts or procedures (Item 8)

Highly relevant

- The problem corresponds directly to familiar concepts:
  - Uniform motion
  - Circular motion
  - Fractions and ratios
- Students can recognise this as:
  - "Find the fraction of a full rotation represented by the given time"

👉 This shows the ability to **map a new situation onto a known mathematical model**

⭐⭐⭐⭐ 4. Identify and describe the mathematical aspects of a real-world problem situation (Item 6)

Important

- Key mathematical aspects include:
  - Time (40 minutes, 30 minutes)
  - Starting position P
  - Direction of rotation (indicated by arrows)
  - Motion along a circular path
- Students must ignore irrelevant information such as height or diameter

👉 This reflects the ability to **extract relevant mathematical variables from a real context**

⭐⭐⭐⭐ 5. Identify the key variables in a model (Item 2)

Important

Key variables include:

- Total time for one rotation
- Elapsed time
- Fraction of the circle completed
- Corresponding position on the circle

👉 These form a **simple time–position model**

⭐⭐⭐ 6. Translate a problem into a standard mathematical representation or algorithm (Item 9)

Moderately relevant

- Students mentally or on paper perform:
  - $\frac{30}{40} = \frac{3}{4}$
  - Equivalent to **270°**
- They then relate this to the diagram

👉 This involves translating **context → mathematics → context**

⭐⭐⭐ 7. Simplify or decompose a situation or problem (Item 7)

Moderately relevant

- Students simplify the task by:
  - Ignoring unnecessary numerical information (e.g. heights, diameter)
  - Reducing the problem to "what fraction of a full circle has been completed?"

👉 This lowers the cognitive load of the situation

⭐⭐⭐ 8. Identify the constraints, assumptions, and simplifications in a mathematical model (Item 12)

Moderately relevant

Implicit assumptions include:

- Constant rotational speed
- Continuous motion without stopping
- The rider always remains on the circular path
- The diagram accurately represents the positions

👉 These are standard **modelling assumptions**

⭐ ⭐ 9. Select a mathematical description or representation that describes a problem (Item 1)

Lower relevance

- Possible representations include:
  - Fractions of a rotation
  - Angles
  - Relative positions on the circle
- However, the problem does not require students to explicitly choose among representations

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⭐ ⭐ 10. Select a representation appropriate to the problem context (Item 3)

Lower relevance

- The representation is largely predetermined by the diagram and context

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⭐ ⭐ 11. Use mathematical tools to describe mathematical structures or relationships (Items 10 & 11)

Lower relevance

- No formal symbols, diagrams, or technological tools are required
- The task relies mainly on reasoning and spatial interpretation

⭐⭐⭐⭐⭐ 1. Read, decode and make sense of statements…（第4条）

最核心

- 学生必须正确理解：
  - “one full rotation in 40 minutes” 表示 **匀速**
  - “half an hour” = **30 minutes**
  - 起点是 **P**
  - 选项对应的是 **图中的位置关系**
- 如果不能准确解读文字和图像信息，就无法判断位置

👉 这是**从文字与图像共同构建情境模型**的关键能力

⭐⭐⭐⭐⭐ 2. Recognise mathematical structure（第5条）

非常关键

- 学生需要识别的数学结构：
  - 时间与转动角度成 **正比例**
  - 40 分钟 → 360°
  - 30 分钟 → $\frac{30}{40} = \frac{3}{4}$ 圈
- 这是一个**周期性、比例性结构**

👉 属于对**周期运动与比例关系**的识别

⭐⭐⭐⭐ 3. Recognise aspects corresponding with known concepts（第8条）

高度相关

- 该问题与以下熟悉概念直接对应：
  - 匀速运动
  - 圆周运动
  - 分数 / 比例
- 学生可将其识别为：
  - “时间占一圈的几分之几 → 位置判断”

👉 体现了**将新情境映射到已知数学模型**

⭐⭐⭐⭐ 4. Identify and describe the mathematical aspects of a real-world situation（第6条）

重要

- 关键数学要素包括：
  - 时间（40 分钟、30 分钟）
  - 起始位置 P
  - 运动方向（由箭头给出）
  - 圆周路径
- 学生需忽略与本问无关的高度、半径信息

👉 是**从现实问题中筛选数学变量**的体现

⭐⭐⭐⭐ 5. Identify the key variables in a model（第2条）

重要

关键变量包括：

- 总转动时间
- 已经过的时间
- 转过的圆周比例
- 对应的空间位置

👉 构成了一个**简单的时间–位置模型**

⭐⭐⭐ 6. Translate a problem into a standard mathematical representation（第9条）

中等相关

- 学生会在头脑中或草稿中完成：
  - $\frac{30}{40} = \frac{3}{4}$
  - 即转过 **270°**
- 再将角度对应到图中位置

👉 是从情境 → 数学 → 情境的双向翻译

⭐⭐⭐ 7. Simplify or decompose the situation（第7条）

中等相关

- 学生需要：
  - 忽略不必要的数值（如高度、直径）
  - 把问题简化为“转了几分之几圈”

👉 有效降低问题复杂度

⭐⭐⭐ 8. Identify constraints, assumptions, simplifications（第12条）

中等相关

隐含假设包括：

- 转速恒定
- 转动连续、无停顿
- 乘客始终在圆周上
- 图中位置标注是精确的

👉 属于**建模中的默认假设**

⭐⭐ 9. Select a mathematical description or representation（第1条）

较低相关

- 可以用：
  - 分数（3/4 圈）
  - 角度（270°）
  - 圆周位置
- 但题目并未要求学生在多种表示间明确选择

⭐⭐ 10. Select a representation appropriate to the problem context（第3条）

较低相关

- 最自然的表示方式已经由题目和图像限定

⭐⭐ 11. Use mathematical tools to describe structures（第10 / 11条）

较低相关

- 不需要正式符号、变量或图形构建
- 更多是心算与空间判断